Cooper Union Satellite Launch Initiative

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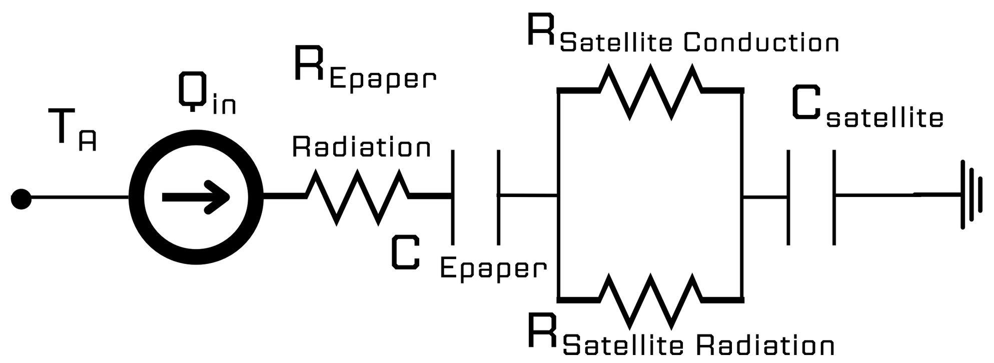
Payload Experimentation

Satellites in low earth orbit (LEO) can go from an extremely cold environment (-65°C) to an extremely hot one (125°C) within the course of 4 hours.

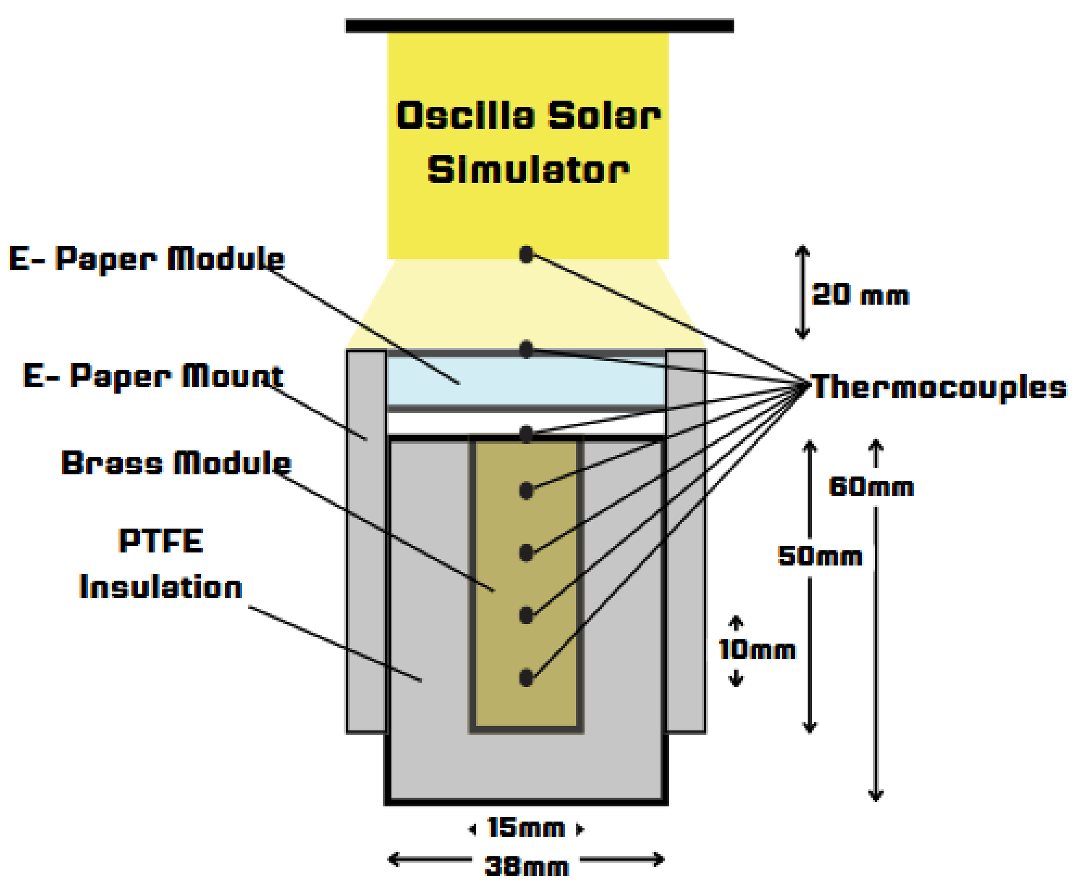
For electronics that must operate at consistent temperatures and are currently heated only by other electronics (resistive heating), thermal regulation can be a high consumer of energy. E-paper is an electrophoretic display that draws a near zero current when the display is not being changed. Because of this power efficiency setting, it is a compelling case study for this experiment.

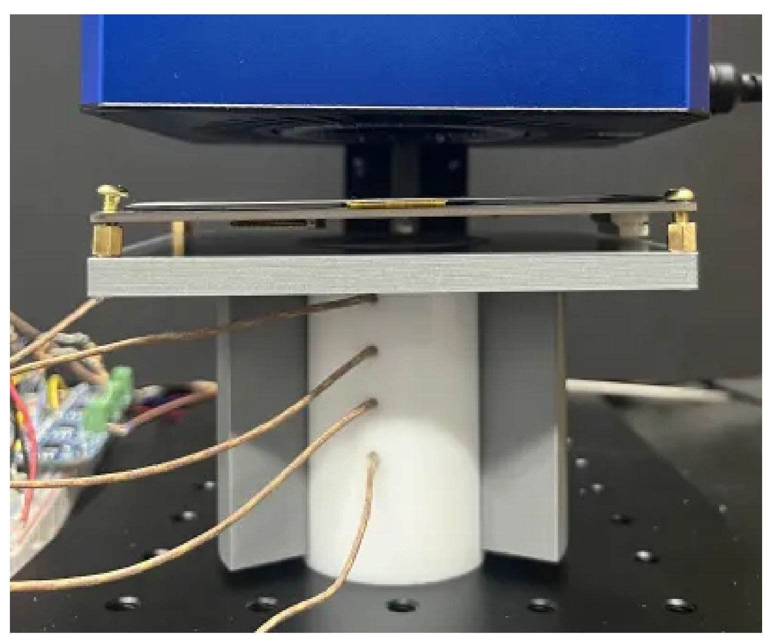
System Simplification: R indicates a thermal resistance, T indicates a temperature, Q indicates a heat flow, and C indicates a thermal capacitance.



$$\frac{-kA_{surface}\Delta T}{\Delta z} = \epsilon\sigma\left(T_{surface}^4 - T_{source}^4\right)$$

The rate of heat flow within the psuedo-satellite is equal to that of the e-paper is given above, where ε is emissivity, σ is the Stefan Boltzman constant, Δz is the distance between the temperature measurements within the psuedo- satellite, and k is thermal conductivity.





An Oscilla LED Solar Simulator is used to mimic sunlight in a controlled environment, producing a near perfect spectral match to the sun.

Type- K thermocouples are used to take both internal and surface temperatures, as they perform well in high temperature environments, including that of LEO.

The light is turned on and remains untouched until steady state is reached. Steady state is determined by calculating the thermal conductivity of brass using the internal temperature measurements and Fourier's law of conductivity, and comparing that to the literature value for that of brass, 115 W/m-K.

Test Setup	Experimental emissivity	Standard deviation
Fully uncolored e-paper	0.0102	±0.00243
Fully colored e-paper	0.0162	±0.133

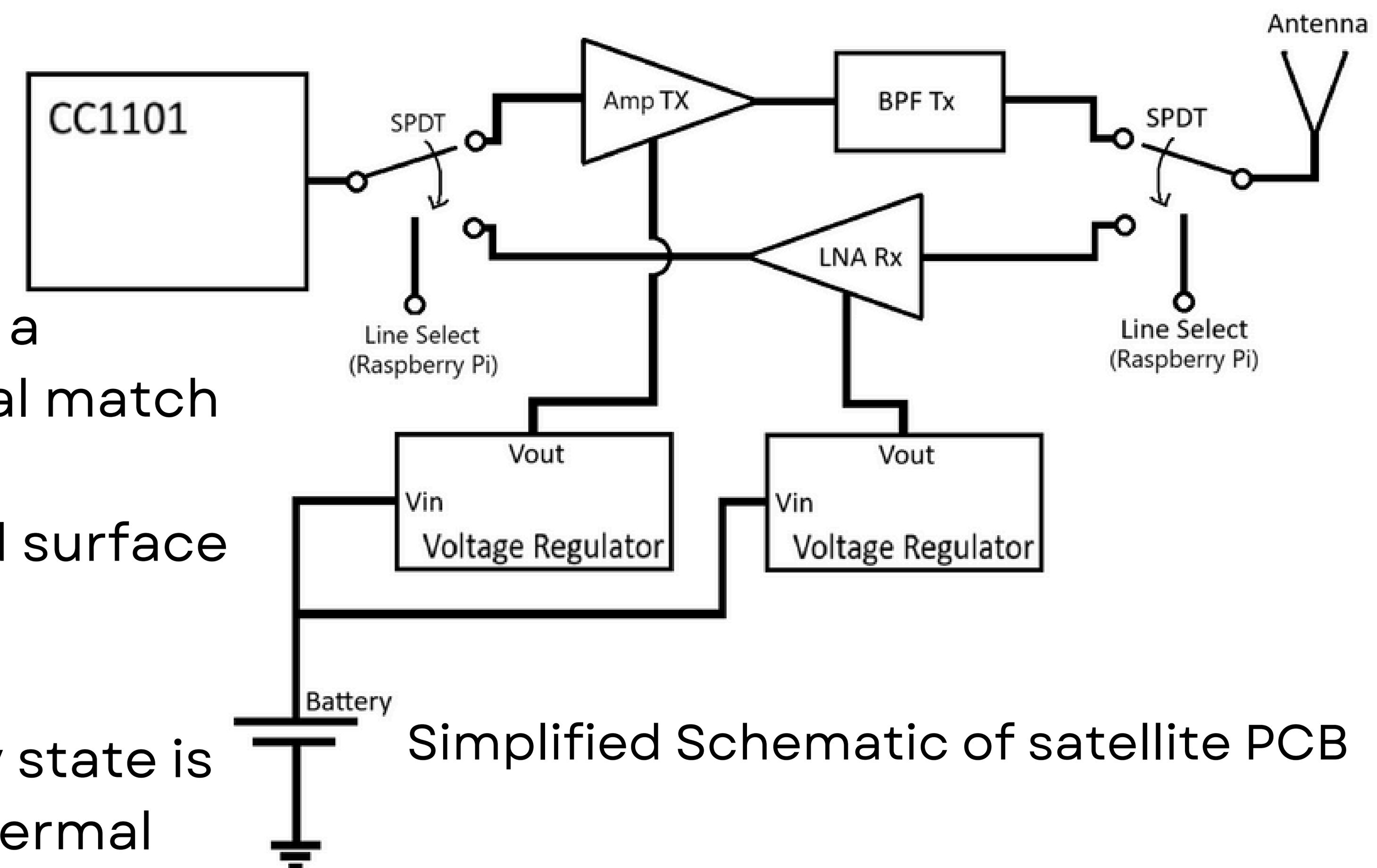
Results for emissivity of e-paper, no vacuum.

Communications Engineering

Communications Engineering is all about making sure that information can be reliably sent from one place to another. For CUSLI, that means controlling the satellite from Earth, and receiving experimental data from space.

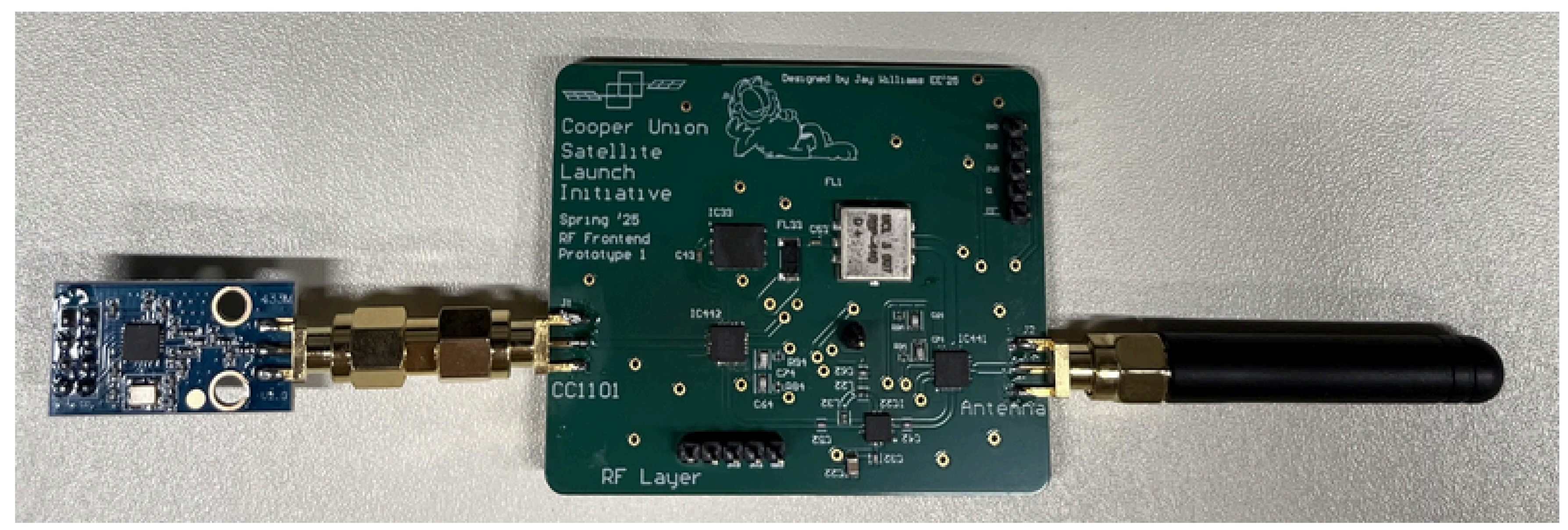
Our satellite's radio is a TI CC1101 RF System-on-chip (RFSoC), and the ground station radio is an Ettus N200 USRP Software-Defined Radio (SDR). With some additional front-end hardware on the satellite side, long distance communication is possible.

Bidirectional communications link between Earth and Space. The ground uses an SDR with a log-periodic antenna.



Simplified Schematic of satellite PCB

Due to size and weight constraints, the satellite's hardware will be built on a printed circuit board, rather than the large coaxially connected components as are seen on the ground station hardware.



1st Prototype of RF Frontend PCB. CC1101 RFSoC on left, antenna on right.